

KEFLAVIK INTL, Iceland

WMO: 040180

Lat: 63.975N

Lon: 22.588W

Elev: 47

StdP: 100.76

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 16201

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-7.2	-6.0	-14.1	1.1	-6.1	-12.1	1.3	-4.8	22.4	2.4	20.4	2.3	5.6	10	0.888

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.6	15.9	11.9	14.8	11.3	13.9	11.0	13.0	14.5	12.3	13.6	11.7	13.0	5.0	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
12.2	8.9	13.1	11.8	8.6	12.8	11.0	8.2	12.1	36.7	14.3	35.0	13.5	33.6	13.0	17.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.3	16.0	13.9	-9.7	18.8	1.8	1.9	-11.1	20.2	-12.2	21.3	-13.2	16.1	-14.1	16.8	-14.5	23.8
(5)			-11.0	14.6	1.7	1.2	-12.2	15.4	-13.2	16.1	-14.1	16.8	-15.4	17.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.2	0.9	0.6	1.3	3.7	6.5	9.6	11.3	10.9	8.5	5.0	2.5	1.1		
	DBStd	4.73	3.20	3.35	3.28	2.85	2.23	1.72	1.49	1.59	2.21	2.78	3.14	3.41		
	HDD10.0	1861	281	262	270	189	110	27	4	8	53	157	225	275		
	HDD18.3	4798	539	495	528	439	368	263	218	231	294	414	475	533		
	CDD10.0	103	0	0	0	0	1	14	44	35	8	1	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	1	0	0	0	0	0	0	0	1	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	6.6	7.4	7.7	7.4	6.8	6.1	5.4	5.3	5.5	6.6	6.6	7.2	7.4		
(15) Precipitation	PrecAvg	1099	103	97	99	77	65	63	67	95	111	115	107	115		
	PrecMax	2362	274	208	229	227	155	148	139	302	280	310	256	202		
	PrecMin	698	22	14	5	19	14	3	15	15	33	16	27	37		
	PrecStd	241	44	43	43	38	29	33	32	57	52	56	47	43		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.2	8.4	9.0	11.1	13.8	16.2	18.0	17.8	14.2	11.9	10.0	9.2		
		MCWB	7.7	8.0	8.1	7.4	9.0	10.8	13.1	13.8	11.8	10.4	9.3	8.8		
	2%	DB	7.4	7.7	8.0	9.2	12.0	14.4	16.1	15.2	13.1	10.8	9.0	8.2		
		MCWB	6.9	7.2	7.3	7.2	8.2	10.3	12.0	12.2	11.1	9.4	8.4	7.7		
	5%	DB	6.8	6.9	7.1	8.5	10.9	13.2	15.0	14.1	12.2	9.9	8.1	7.2		
		MCWB	6.3	6.3	6.3	6.9	7.8	9.7	11.4	11.5	10.6	8.9	7.5	6.7		
	10%	DB	5.8	5.8	6.1	7.8	9.8	12.3	14.0	13.2	11.7	9.0	7.1	6.2		
		MCWB	4.9	5.0	5.0	6.5	7.5	9.4	11.1	11.0	10.3	8.2	6.2	5.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.1	8.2	8.2	8.6	10.0	12.2	14.0	14.6	12.7	11.1	9.5	8.7		
		MCDB	8.1	8.2	8.4	9.2	11.9	14.2	16.5	16.7	13.4	11.4	9.6	8.8		
	2%	WB	7.0	7.1	7.4	7.9	9.2	11.3	12.9	13.1	11.9	9.8	8.5	7.7		
		MCDB	7.2	7.3	7.7	8.7	10.5	13.1	14.8	14.5	12.5	10.2	8.7	8.0		
	5%	WB	6.1	6.1	6.4	7.2	8.6	10.7	12.3	12.4	11.2	9.0	7.4	6.7		
		MCDB	6.5	6.5	6.9	8.0	9.8	12.4	13.9	13.6	11.9	9.5	7.9	7.1		
	10%	WB	4.6	4.8	5.1	6.5	8.0	10.1	11.7	11.7	10.6	8.1	6.2	5.4		
		MCDB	5.4	5.4	5.9	7.4	9.1	11.8	13.3	12.9	11.5	8.9	6.8	6.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.1	4.4	4.5	4.9	5.0	4.8	4.6	4.4	4.2	4.0	4.0	4.0		
		MCDBR	4.6	4.7	4.6	4.9	6.0	6.2	6.5	5.5	4.4	4.1	4.2	4.4		
		MCWBR	5.1	4.9	4.5	4.0	3.8	3.6	3.7	3.5	3.5	4.0	4.7	4.9		
		MCDBR	4.6	4.7	4.4	4.5	4.7	5.3	5.1	4.5	3.7	4.0	4.3	4.5		
	5% WB	MCWBR	5.2	5.1	4.8	4.2	3.6	3.6	3.5	3.4	3.7	4.3	4.9	5.1		
(40) Clear-Sky Solar Irradiance	taub		0.409	0.313	0.338	0.363	0.361	0.350	0.351	0.346	0.343	0.306	0.367	0.371		
	taud		2.541	2.240	2.330	2.323	2.396	2.480	2.503	2.527	2.452	2.326	2.672	2.588		
	Ebn at Noon		277	596	748	810	850	870	858	830	745	602	332	158		
	Edh at Noon		30	58	80	100	101	95	91	81	71	52	25	16		
(44) All-Sky Solar Radiation	RadAvg		0.12	0.53	1.52	3.21	4.76	5.13	4.67	3.58	2.07	0.87	0.19	0.05		
	RadStd		0.01	0.08	0.21	0.38	0.56	0.71	0.47	0.28	0.20	0.07	0.02	0.00		

Historical Trends

		Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
(47) Regional (7 neighbors)	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
		+0.39	+0.91	+1.49	N/A	N/A	N/A	-123	-141
		+0.53	+1.04	+1.16	+0.72	N/A	N/A	-160	-193

Nomenclature: See separate page